Business Analysis Report

Executive Summary

Project Overview

This project analyzes the Global Superstore dataset, containing 51,290 transactional records across 27 fields from 2011-2014 to evaluate sales performance, profitability, product performance and data quality. The analysis uses SQL and Python, with Tableau visualization to communicate key business findings.

The dataset generated approximately $12.64 million in sales, and generated $1.47 million in profit, and an overall 11.61% profit margin.

Technology had the highest sales among the three categories, at approximately $4.74 million, and the highest category-level profit margin at 13.99%. Furniture generated about $4.11 million in sales but had a lower profit margin of 6.94%, indicating an opportunity for further profitability analysis.

The Tables sub-category generated approximately $757,034 in sales but recorded a loss of $64,083.39, resulting in a profit margin of -8.47%. Product-level analysis identified 210 unprofitable products of 366 distinct products in the Tables sub-category. These products collectively recorded a loss of $117,166, exceeding the approximately $53,083 in profit generated by profitable products.

Discount analysis also identified a difference in average product-level discounts: 40.15% for unprofitable products compared with 15.01% for profitable products in the Tables sub-category. This is a descriptive association and does not establish that discounts caused the losses.

The data quality assessment identified one zero-sales record, 12,544 records with negative profits, and 10,378 records where Market and Market2 value differed. These findings require proper business interpretation before any operating or business decisions are made. No negative-sales records or invalid shipping-date sequence were identified in the checks performed.

Overall, the findings suggest that the Tables sub-category warrants further profitability review and that data definition should be clarified to support reliable reporting. Customer-segment profitability analysis and product rankings have been completed, and their findings are included in this report.

Key Findings

1. **Overall Business Performance:**

   The dataset recorded approximately $12.64 million in sales, $1.47 million in profit, resulting in an overall profit margin of 11.61%.

2. **Category-Level Performance:**

   Technology was the strongest-performing category, generating approximately $4.74 million in sales, $664,000 in profit, and a 13.99% profit margin. Office Supplies followed with a profit margin of 13.69%, while Furniture had the lowest profit margin of 6.94%.



3. **Tables Sub-Category Profitability:**

   The Tables sub-category generated approximately $757,034 in sales but recorded a loss of $64,083.39, resulting in a profit margin of -8.47%. This indicates that high sales do not necessarily translate into profitability.

4. **Product-Level Profitability:**

   Of the 366 distinct products, 210 were classified as unprofitable, 155 as profitable, and one as break-even. The unprofitable products recorded approximately $117,166 in losses, while profitable products generated approximately $53,083 in profit. The combined losses exceeded the gains from the profitable products.

5. **Discount and Profitability Association:**

   The average product-level discount was 40.15% for unprofitable products in the Tables sub-category and 15.01% for profitable products in the Tables sub-category, a difference

of 25.14 percentage points. This indicates an association between discount levels and product profitability in this analysis, but does not establish causation.

6. **Shipping-Cost and Profitability Analysis: Tables Sub-Category**

   The Tables sub-category had negative profit margins across the shipping-cost bands below $200, while the $200-and-above band had a positive profit margin of 6.10%. These descriptive results do not establish that shipping cost caused the differences in profit.

7. **Data Quality Findings:**

   The data quality assessment identified one record with zero sales, 12,544 records with negative profit, and 10,378 records where the Market and Market2 values differed. These findings require business interpretation. The checks also found no negative-sales records, no invalid shipping-date sequences, no missing values in the selected key fields, and no duplicate Row.ID values.

8. **Top Products by Sales and Profit:**

   The Canon image CLASS 2200 Advanced Copier (Product ID: TEC-CO-10004722) ranks first in both total sales and total profit. Across five order lines, the product generated $61,600 in total sales and $25,199.93 in total profit, representing an aggregate profit margin of 40.91%. This finding identifies the product as a strong contributor to the overall sales and profitability in the historical dataset. However, its performance alone does not establish future demand or guarantee similar profitability.

9. **Customer Segment Profitability:**

   The Consumer segment generated the highest total profit, at $749,239.78, from approximately $6.51 million in sales, with a profit margin of 11.51%. The Corporate segment generated $441,208.33 in profit, while the Home Office generated $277,009.18.

Although Consumer generated the highest total profit, Home Office had the highest profit margin at 11.99%, compared to 11.54% for Corporate and 11.51% for Consumer. This indicates that the Consumer segment contributes the most profit in absolute terms, while the Home Office segment generates slightly more profit per dollar of sales.

These results distinguish overall profit contribution from profit efficiency and help management evaluate customer segments using both measures.

10. **Underperforming Products by Total Profit:**

A product-level analysis identified the 10 products with the lowest cumulative profit across the dataset. The Cubify CubeX 3D Printer Double Head Print recorded the largest loss among these products, at $8,879.97 across three order lines. Other notable loss-making products included the Hoover Stove, White, with a loss of $4,958.16, and the Apple Smart Phone, Full Size, with a loss of $4,574.64.

The Lesro Computer Table, Fully Assembled, had a particularly negative profit margin of -233.21%, recording a loss of $2,798.49 on $1,200 in sales across three order lines. This illustrates that the product with the worst profit margin is not necessarily the product with the largest total loss.

These results identify products for further profitability review. Before considering pricing changes or discontinuation, management should examine transaction-level discounts, costs, pricing, and the amount of available sales history. The analysis is descriptive and does not establish the causes of the losses.

11. **Shipping-Cost and Profitability Analysis: Full Dataset**

An analysis of the full Global Superstore dataset found that profit margin increased across the shipping-cost bands. The under-$25 shipping-cost band had a profit margin of

7.81%, while the $200-and-above shipping-cost band had the highest profit margin of 17.06% and the highest total profit of $376,196.12.

The observed pattern indicates an association between shipping-cost bands and profitability in this dataset. However, the analysis does not establish that higher shipping cost causes higher profits. The groups differ in transaction volume and may also differ in product mix, order values, and other factors. Further investigation would be needed before making operational decisions based on shipping costs.

12. **Discount and Profit Analysis:**

An analysis of discount levels across the full Global Superstore dataset found a pattern of declining profitability as discount increased. Transactions with no discount generated $1,828,671.85 in total profit with a profit margin of 25.21%. At a 10% discount, the profit margin was 16.40%, while at 25%, the profit margin fell to 0.91%; at 27% discount, the profit margin became negative at -1.87%. As the discount increased at 30% and above, the aggregated results showed a negative profit margin. For instance, the 40% discount group recorded a loss of $155,178.90 and profit margin of -26.16%, while the 50% discount group recorded a loss of $158,629.88 and profit margin of -48.94%.

These findings indicate an association between higher discount levels and lower profitability. However, the analysis does not establish that discount alone caused the loss. Product mix, costs, and other transactional characteristics may also affect profitability. Management should investigate high-discount transactions and review their profitability before implementing any discount changes or policies.

13. **Average Order Value (AOV):**

The dataset consists of 25,035 distinct orders and $12,642,905 in total sales, resulting in an average order value of $505.01. This metric represents the average sales revenue per order across the 2011-2014 dataset.

14. **Average Shipping Time:**

    The average elapsed time between order date and ship date was 3.97 calendar days, approximately 4 days, across the dataset's order lines. This metric provides a baseline for understanding historical shipping turnaround time in the 2011-2014 dataset.

Limitations

This analysis is descriptive and uses historical data from 2011-2014. It identifies patterns and potential areas for investigation but does not establish causation. Some products have very few transaction lines, and the dataset does not provide a full business context behind pricing, discounts or costs. Recommendations should therefore be validated with business stakeholders before operational changes are implemented.

Project Status

The initial SQL business analysis, data quality assessment, product-level profitability investigation, and supporting business analyses have been completed. The findings are documented in this report and visualized in the Tableau dashboard.

Completed Deliverables

- Business Requirements Document (BRD) — Version 1.0
- Data Quality Test Cases (DQ-001 through DQ-008)
- Initial SQL Business Analysis
- Tables Sub-category Product-Level Profitability Investigation
- Product Performance Rankings

- Customer-Segment Profitability Analysis
- Shipping-Cost and Profitability Analysis
- Discount and Profitability Analysis
- Initial Business Recommendations
- Tableau Dashboard

Potential Next Steps

- Complete a final consistency review of the report, SQL results, and Tableau dashboard.
- Publish and organize the project artifacts in the portfolio repository.
- Document any remaining limitations or validation tasks.

Business Recommendations

Based on the previous analysis of sales, profitability, discount, and product performance, the following recommendations are proposed for further business investigation.

1. **Review Discounting Practices for the Tables Sub-Category**

   Products in the Tables sub-category with negative cumulative profit had a higher average product-level discount (40.15%) than profitable products (15.01%). Management should review high-discount transactions and evaluate whether discount approval thresholds or minimum-margin checks are appropriate. These findings show an association between discount levels and profitability but do not establish that discount alone caused the losses.

2. **Prioritize Unprofitable Products in the Tables Sub-Category**

   Of 366 distinct products in the Tables sub-category, 210 had a negative cumulative profit. The combined losses for these products exceeded the profits generated by the profitable products. Management should prioritize products with substantial total losses and

sufficient transaction history for meaningful reviews. Management should examine pricing, product costs, discount patterns, and shipping expenses before considering pricing changes or product discontinuation.

3. **Establish Ongoing Profitability Monitoring**

   Develop a recurring report or dashboard tracking profit, sales, profit margin, discount levels and product-level performance. This would help management identify loss-making products and investigate emerging profitability issues earlier.

4. **Clarify Geographic Market Definitions**

   The dataset consists of 10,378 records where Market and Market2 fields differ. The data owner should confirm or clarify whether these fields represent different levels of geographic classification and document their intended definitions and relationship to improve reporting consistency.

5. **Validate the Zero-Sales Transaction**

   One record contains $0 in sales and negative profit of $1.11. Confirm whether it represents a valid adjustment, replacement, promotion or other business scenario. Document the business decision before changing or excluding the record from reporting.